

Test 3

March 17, 2024

Problem 1

Let $\triangle ABC \sim \triangle DEF$ and let $\overline{AB} = 3$, $\overline{AC} = 4$, and $\overline{BC} = 5$. Find the lengths $\overline{DE}$ and $\overline{DF}$ given that:

a) $\overline{EF} = 10$

b) $\overline{EF} = 25$

c) $\overline{EF} = 2.5$

Problem 2

Prove that if the legs of one right triangle are proportional to the legs of another right triangle, then the triangles are similar.

Problem 3

Prove that if two triangles are similar, then the ratio of their perimeters is the same as the ratio of two corresponding sides.

Problem 4

Two triangles have the same side for their bases. The height of one exceeds that of the other by 4 *in*. The areas of the two triangles differ by 40 *in*². What is the length of the common base of the two triangles?

Problem 5

A trapezoid has a height of 6 in and an area of 39 in^2 . If one of the bases has a length of 6 in , what is the length of the other base?

Problem 6

Find the radius and apothem of an equilateral triangle whose side lengths are 2 in .

Problem 7

On a circle of radius 2 is an arc of length 4. How many radians are there in the central angle?

Problem 8

A triangle has angles of measures 30° , 60° , and 90° . Given that the hypotenuse of the triangle has a length of 1, find the lengths of the legs.

Problem 9

Let the square $\square ABCD$ be inscribed by circle C_1 , and circumscribe circle C_2 . Prove that the area of C_1 is double that of C_2 .

Extra Credit

Prove the Pythagorean Theorem. That is, prove that the square of the length of the hypotenuse of a right triangle is equal to the sum of the squares of the lengths of the legs.